

This is a high-octane sprint. With a **9:00 PM** hard deadline and a 75-minute window remaining, we need to strip away any non-essential complexity and focus on a "vertical slice" that is playable, visually striking, and scientifically grounded.

Here is the final, unified blueprint for **Escape from Mars**.

1. Core Mission Objective

Launch the Starship from the Jezero Crater before the **Trans-Earth Injection** window closes. You must manage your body's physical decay and the ship's fuel production through automated systems before you reach a lethal dose of radiation or become too weak to fly.

2. Rapid-Fire Team Assignments

Michael |

Systems & UX Lead

- **The Sabatier Console:** Create the UI for the reactor to convert $\text{CO}_2 + 4\text{H}_2 \rightarrow \text{CH}_4 + 2\text{H}_2\text{O}$.
- **Mass-Management Inventory:** Implement the trade-off system where carrying scientific samples increases the fuel requirement for launch.
- **The Biological Clock:** Code the ticking "Death Clock" for **Bone Density** and **Radiation Exposure**.
 - *Learning Hook:* Bone density drops ~1% per month in low-G environments.

Evan |

Mechanics & Physics Lead

- **38% Gravity Controller:** Set the gravity constant to 3.721 m/s^2 . Jumps should feel floaty and slow (2.6x higher than Earth).
- **The Window Timer:** A global countdown to the launch window. If the timer hits zero before the fuel bar is full, the mission fails.
- **Launch Logic:** Build the 3-stage ignition sequence: **Power $\rightarrow$ Prime $\rightarrow$ Launch**.

Roberto |

Environment & Aesthetics Lead

- **The Martian Landscape:** Build the Jezero Crater using "Leopard Spot" rock textures and a Stainless Steel Starship model.
- **Atmospheric Cues:** Implement the **Blue Sunset** (Rayleigh scattering) as the visual indicator that the window is closing.

- **Audio:** Apply a "muffling" filter to all sounds to simulate the thin Martian atmosphere.

3.

Core Gameplay Loop (The "Must-Haves")

Feature	Physics/Chemistry Basis	Gameplay Impact
Sabatier Fueling	In-Situ Resource Utilization (ISRU).	Player clicks "Process" to convert H_2 and CO_2 into Methane.
Dust Maintenance	Electrostatic dust buildup.	Solar power drops over time; player must "brush" panels to keep the factory powered.
Low-G Movement	3.721 m/s^2 gravity.	High jumps are possible, but fall damage triggers at greater heights.
Payload Stress	Tsiolkovsky Rocket Equation.	Every 1kg of "Science Samples" added to inventory increases the Methane requirement.

4. The 75-Minute "Crunch" Schedule

Time Window	Phase	Task

7:45 - 8:00	Foundation	Michael sets up the GameState (Health, Fuel, Power). Evan implements the Gravity jump. Roberto places the ship model.
8:00 - 8:30	Integration	Michael connects the Sabatier UI to the Fuel bar. Evan links the "Launch" button to the fuel requirement. Roberto adds the Blue Sunset.
8:30 - 8:50	The Merge	Combine all scripts. Test the "Game Over" states (Radiation death vs. Missing the window).
8:50 - 9:00	Deployment	Final build export. No new code. Check the HUD text for clarity.

5. Technical Constants

- **Gravity:** 3.721 m/s^2 .
- **Fuel Chemistry:** $\text{CO}_2 + 4\text{H}_2 \rightarrow \text{CH}_4 + 2\text{H}_2\text{O}$.
- **Delta-v Goal:** $\Delta v = v_e \ln \frac{m_0}{m_f}$.

Let's get to work. Michael, do you want me to generate the logic for the Sabatier UI conversion or the Bio-Decay timer first to get your code block started?